ORF307
Integer Optimization

Ioannis Akrotirianakis

Announcements

The Logistics of the final exam:

- The final exam will be a **36-hour take-home assessment**.
- The exam problems will be available for download on Gradescope **starting December 13th at 7:00 AM EST**.
- The deadline for submitting your solutions on Gradescope is **December 14th at 7:00 PM EST**.
- Solutions submitted after this deadline will not be accepted. Please ensure you have sufficient time to create and **upload the final PDF file of your solutions.** And remember you can *resubmit* the PDF with your solutions as many times as you want until the time expires.
- **Additional information will be posted on Canvas and EdDiscussion shortly!**

Announcements

- Last precepts this week → please attend
- HW8 out Monday (yesterday) → deadline **Thursday December 11, 11:59PM**

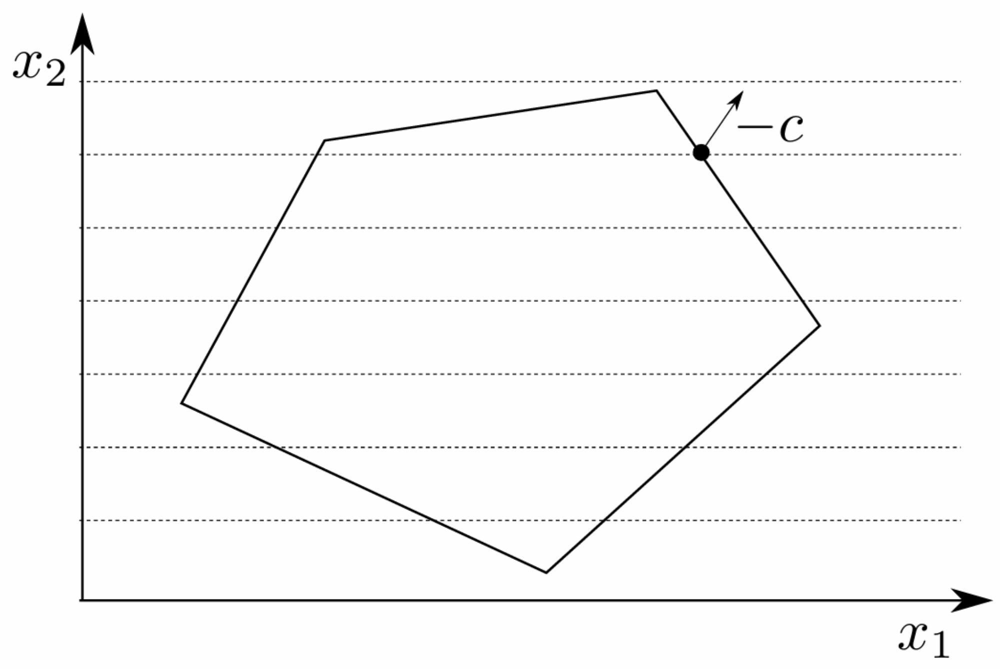
Today's topics

- Mixed Integer Programs (MIP)
- Modeling techniques
- Formulations
- Ideal formulations

- **Mixed Integer Optimization**

Mixed integer program (MIP)

Definition: a MIP is an optimization problem where some of the decision variables must take integer values only!



$x_1 \in R$ and $x_2 \in Z$

Mathematical formulation:

$$\min \; c^T x$$
$$\text{s.t.} \quad Ax \le b$$
$$x_i \in Z, \; i \in I$$

where Z is the set of integer numbers, and $\boldsymbol{I} \subseteq \{\mathbf{1}, \mathbf{2} \ldots, \boldsymbol{n}\}$

Feasible region: dotted lines within the polytope

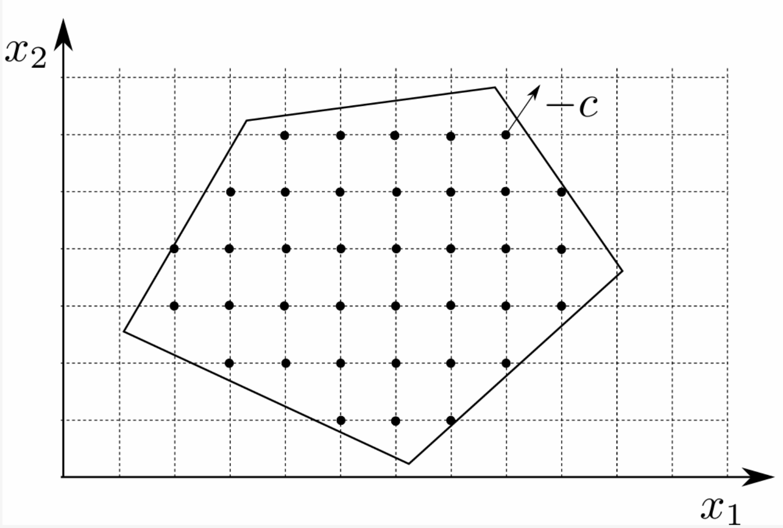
Mixed integer program (MIP) – Special cases

Integer Linear Program (ILP)

$$\min\ c^T x$$

$$\text{s.t.}\quad Ax \leq b$$

$$x_i \in Z,\ \ i \in I = \{1, \ldots, n\}$$

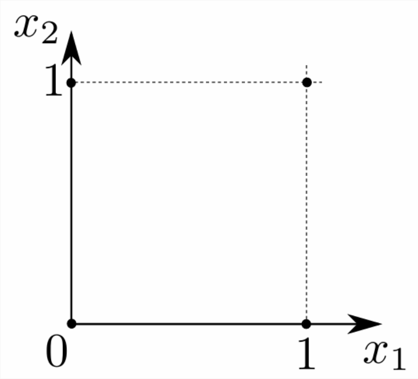


Boolean Linear Program

$$\min\ c^T x$$

$$\text{s.t.}\quad Ax \leq b$$

$$x_i \in \{0,1\},\ \ i \in I$$



- **Modeling techniques**

Binary choice

$$x_i = \begin{cases} 1, \text{ if event occurs} \\ 0, \text{ otherwise} \end{cases} \longrightarrow \quad x_i \in \{0,1\} \quad \text{and} \quad x \in \{0,1\}^n$$

Examples:

- Perform a financial transaction (or not)
- Select a node (or an arc) in a graph (or not)
- Open a store (or not)

Knapsack problem

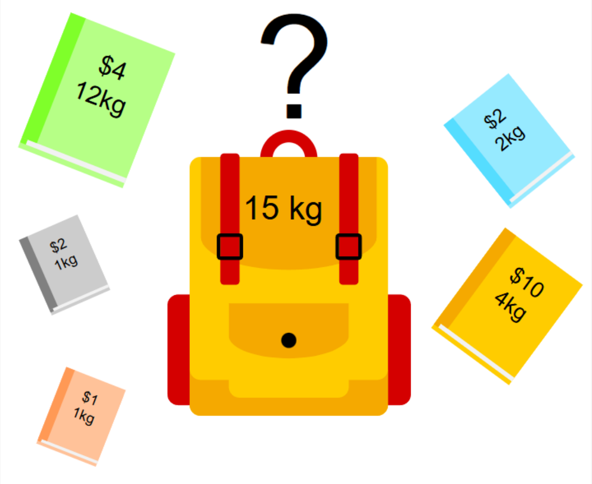
Goal: decide what items to put into your knapsack

Data

- Maximum total weight your knapsack can hold
- Weight of the i-th item: a_i
- Value of the i-th item: c_i
- There are n items for you to select
- Maximum weight your knapsack can hold: b

Decision variables:

- x_i: **select** the i-th item (**or not**), $i \in \{1, \dots, n\}$



Wikipedia image

Formulation:

$$\min \; c^T x$$

$$\text{s.t.} \quad a^T x \le b$$

$$x_i \in \{0,1\}, \; i \in \{1, \dots, n\}$$

Logical relations

Consider a binary variable: $x \in \{0,1\}$

At most one event occurs:

$$\mathbf{1}^T x = 1$$

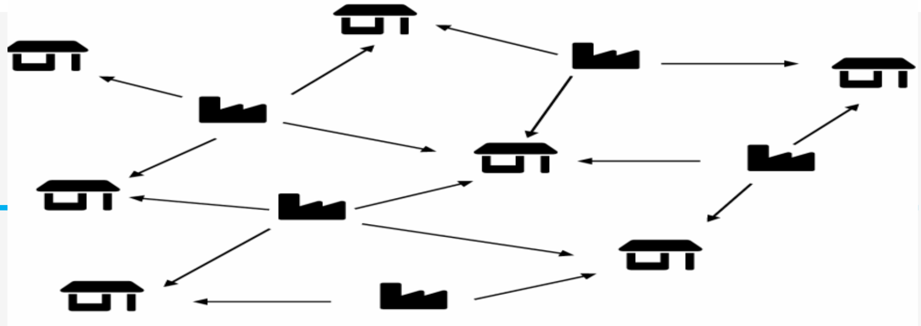
Neither or both events occur:

$$x_1 - x_2 = 0$$

If $x_2 = 0$ (does not occur), then $x_1 = 0$ (does not occur)

$$x_1 \leq x_2$$

Facility location problem



Goal: determine the **best location** of n facilities (e.g., factories or warehouses) based on <u>construction costs</u> and <u>distance to customers</u>

Data:

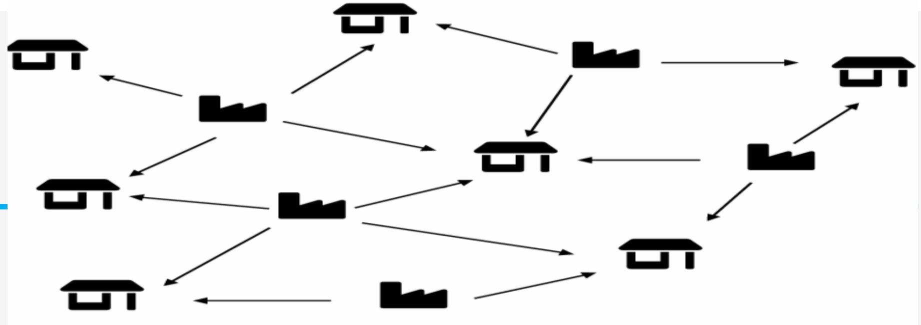
- n: possible facility locations
- m: customers
- c_j: cost of building a facility at location j
- d_{ij}: cost of serving customer i from location j

Constraints:

- A customer can be served by one location only!

Decision variables:

Facility location problem



Goal: determine the **best location** of n facilities (e.g., factories or warehouses) based on construction costs and distance to customers

Data:

- n: possible facility locations
- m: customers
- c_j: cost of building a facility at location j
- d_{ij}: cost of serving customer i from location j

Constraints:

- A customer can be served by one location only!

Decision variables:

$$y_i = \begin{cases} 1, \text{ if location } j \text{ is selected} \\ 0, \text{ otherwise} \end{cases}$$

$$x_{ij} = \begin{cases} 1, \text{ if customer } i \text{ is served from location } j \\ 0, \text{ otherwise} \end{cases}$$

Formulation:

$$\min \sum_{j=1}^{n} c_j y_j + \sum_{i=1}^{m} \sum_{j=1}^{n} d_{ij} x_{ij}$$

$$\text{s.t. } \sum_{j=1}^{n} x_{ij} = 1, \ \ i = 1, \dots, m$$

$$x_{ij} \le y_j, \ i = 1, \dots, m, \ j = 1, \dots, n$$

$$x_{ij}, y_j \in \{0,1\}, \ \ \forall i, j$$

Mixed logical relations (big-M formulation)

Consider a continuous variable $x \in R$ and a binary variable $y \in \{0,1\}$

Logical relation: if $y = 0$ then $x = 0$

Example: if a factory is not constructed ($y = 0$) then it does not produce anything ($x = 0$)

$$0 \leq x \leq yM$$

Disjunctive constraints: <u>either</u> $a_1^T x < b_1$ <u>or</u> $a_2^T x \leq b_2$ <u>hold</u>

$$a_1^T x \leq b_1 + yM$$

$$a_2^T x \leq b_2 + (1 - y)M$$

Cardinality

$x \in R^n$ and $y \in \{0,1\}^n$

Cardinality of a vector $x \in R^n$ is the number of its nonzero elements ($card(x)$)

The **0-norm** of a vector $x \in R^n$ is defined as number of its nonzero elements $(||x||_0)$

$$card(x) = ||x||_0 = \sum \{i \mid x_i \neq 0\}$$

Cardinality constraint:

$$card(x) \leq k \quad \longrightarrow \quad \begin{cases} \sum_{i=1}^{n} y_i \leq k \\ \left.\begin{array}{c} -My_i \leq x_i \leq My_i \\ y_i \in \{0,1\} \end{array}\right\} i = 1, \dots, n \end{cases}$$

Restricted range of values

Goal: restrict variable $x \in R$ to take a value from the set $\{a_1, \dots, a_m\}$ only

Introduce m new binary variables $y_i \in \{0,1\}$, $i = 1, \dots, m$

Define the new constraints:

Only one value can be taken $\longrightarrow$ $x = \sum_{i=1}^{m} a_i y_i$

$$\sum_{i=1}^{m} y_i = 1$$

$$y_i \in \{0,1\}, \forall i$$

Only one y_i can be 1, all others are 0

In vector form:

$$x = a^T y$$

$$\mathbf{1}^T y = 1$$

$$y \in \{0,1\}^m$$

Signal decoding

Message $x \in R^n \longrightarrow \boxed{y = Hx + v} \longrightarrow$ Signal $y \in R^m$

$v \in R^m$ is the vector of **noise or errors**,
and $H \in R^{m \times n}$ (**channel matrix**)

Signal constellation: at every time k, x_k can take values in $\{a_1, \dots, a_d\}$

Goal: recover message $\hat{x}$

Signal decoding problem:

$$\min \ \left|\left|Hx - y\right|\right|_1$$
s.t. $x_k \in \{a_1, \dots, a_d\}$, $k = 1, \dots, n$

Signal decoding as MIP

Signal decoding problem:

$$\min \ \left|\left|Hx - y\right|\right|_1$$
$$\text{s.t. } \ x_k \in \{a_1, \dots, a_d\}, \quad k = 1, \dots, n$$

MIP problem:

$$\begin{aligned} \min \quad & 1^T u \\ \text{s.t.} \quad & -\mathrm{u} \leq \mathrm{Hx} - \mathrm{y} \leq \mathrm{u} \\ & x_k = a^T z_k \\ & 1^T z_k = 1 \qquad , \quad k = 1, \dots, n \\ & z_k \in \{0,1\}^d \end{aligned}$$

Decision variables:

$u \in R^m$

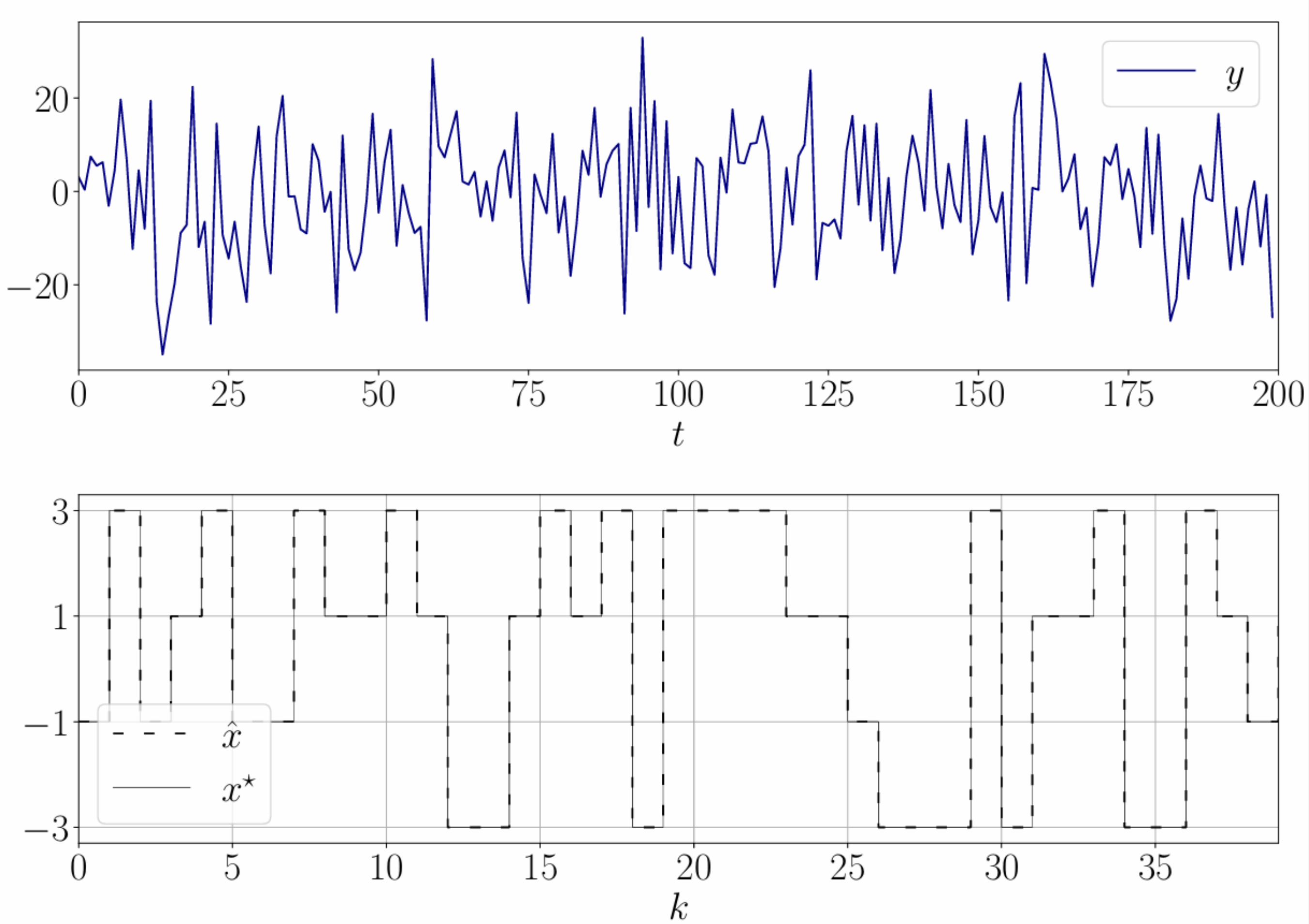
$x_k \in R^n$

$z_k \in \{0,1\}^d$

Signal decoding - Example

Exact message: $\hat{x} \in \{-3, -1, 1, 3\}^{40}$

Noisy signal: $y = H\hat{x} + v \in R^{200}$



Exact message, decoded!

- **Relaxations**

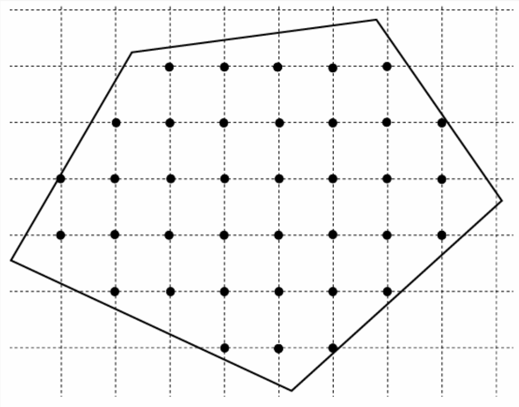
Relaxations

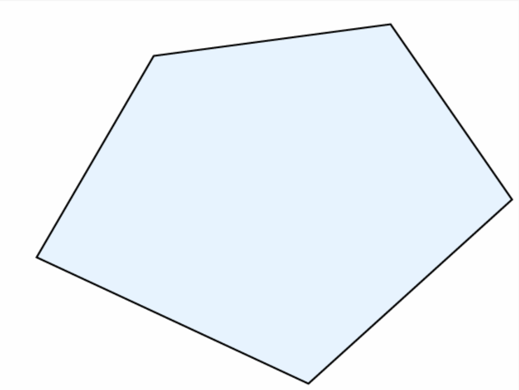
Remove integrality constraints → allow integer variables to take real values

$$\min \; c^T x \quad \text{s.t.} \quad Ax \le b, \quad x_i \in Z, \; i \in \{1, \ldots, n\} \qquad \longrightarrow \qquad \min \; c^T x \quad \text{s.t.} \quad Ax \le b$$

P_{ip} → $x_i \in Z, \; i \in \{1, \ldots, n\}$

P_{rel} → $Ax \le b$





$P_{ip} \subset P_{rel}$ → relaxations provide lower bounds to p_{ip}^*

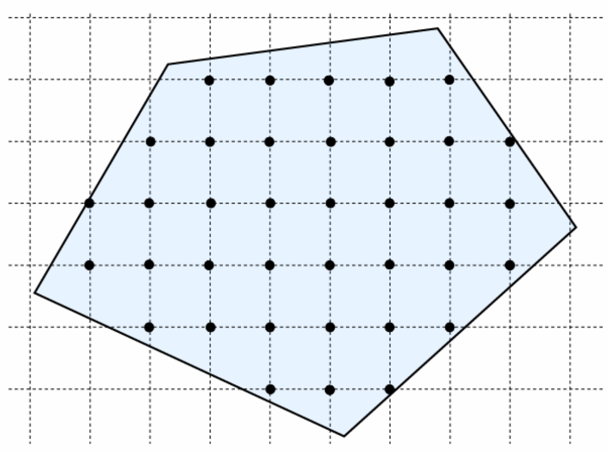
$$p_{ip}^* \le p_{rel}^*$$

Multiple relaxations may exist

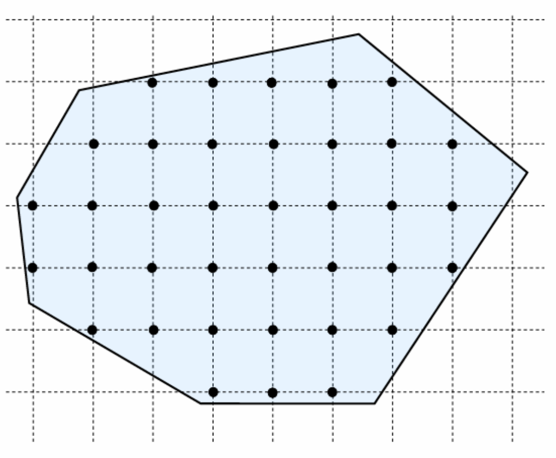
- There may be many equivalent formulations with different relaxations
  - Each formulation contain all the original integer points

$$\begin{aligned} \min\ & c^T x \\ \text{s.t.}\ & Ax \leq b \\ & x_i \in Z,\ i \in I \end{aligned}$$

Formulation 1



Formulation 2

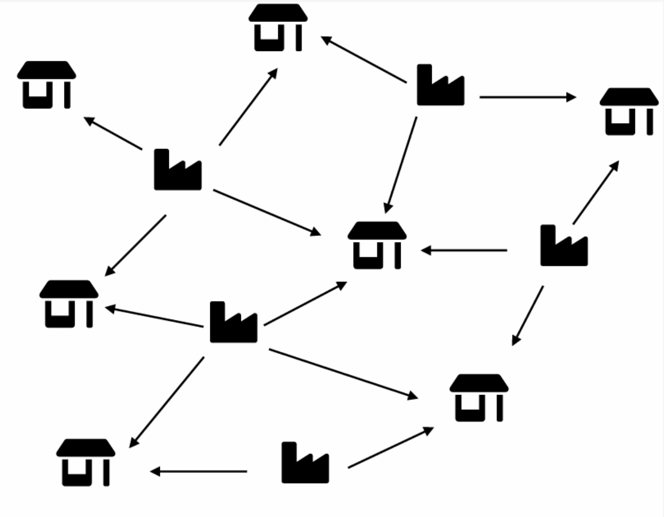


Which one is better?

$$p^*_{rel1} \leq p^*_{rel2}?$$

$$p^*_{rel1} \geq p^*_{rel2}?$$

Facility location problem – Multiple formulations



Formulation 1:

$$\min \sum_{j=1}^{n} c_j y_j + \sum_{i=1}^{m} \sum_{j=1}^{n} d_{ij} x_{ij}$$

$$\text{s.t.} \;\; \sum_{j=1}^{n} x_{ij} = 1, \quad i = 1, \dots, m$$

$$\boldsymbol{x_{ij} \leq y_j, \;\; i = 1, \dots, m, \; j = 1, \dots, n}$$

$$x_{ij}, y_j \in \{0,1\}, \;\; \forall i, j$$

Formulation 2 (fewer constraints):

$$\min \sum_{j=1}^{n} c_j y_j + \sum_{i=1}^{m} \sum_{j=1}^{n} d_{ij} x_{ij}$$

$$\text{s.t.} \;\; \sum_{j=1}^{n} x_{ij} = 1, \quad i = 1, \dots, m$$

$$\boldsymbol{\sum_{i=1}^{m} x_{ij} \leq m y_j, \quad j = 1, \dots, n}$$

$$x_{ij}, y_j \in \{0,1\}, \;\; \forall i, j$$

Decision variables:

$$y_i = \begin{cases} 1, \text{ if location } j \text{ is selected} \\ 0, \text{ otherwise} \end{cases}$$

$$x_{ij} = \begin{cases} 1, \text{ if customer } i \text{ is served from location } j \\ 0, \text{ otherwise} \end{cases}$$

Are the two formulations equivalent?

Which formulation is better?

Facility location problem – Multiple formulations

Formulation 1:

$$P_{rel1} = \left\{ x \mid \sum_{j=1}^{n} x_{ij} = 1, \qquad x_{ij} \leq y_j, \qquad x_{ij}, y_j \in \{0,1\} \right\}$$

Formulation 2:

$$P_{rel2} = \left\{ x \mid \sum_{j=1}^{n} x_{ij} = 1, \qquad \sum_{i=1}^{m} x_{ij} \leq m y_j, \qquad x_{ij}, y_j \in \{0,1\} \right\}$$

Relationship (proof at next slide)

$$P_{rel1} \subset P_{rel2} \quad \Rightarrow \quad p^*_{rel2} \leq p^*_{rel1} \leq p^* = p^*_1 = p^*_2$$

p^*_{rel1} is a larger lower bound and hence Formulation 1 is better

p^*_{rel1} is closer to the optimal solution of the Integer Program p^*

Facility location problem – Multiple formulations

Proof of $P_{rel}1 \subset P_{rel2}$

Formulation 1: P_{rel1}

$$x_{ij} \le y_j, \forall i,j \;\Leftrightarrow\; \max_{\mathrm{i}} \{x_{ij}\} \le y_j, \forall j$$

Formulation 2: P_{rel2}

$$\sum_{i=1}^{m} x_{ij} \le m y_j, \forall j \;\Leftrightarrow\; \underset{i}{\boldsymbol{avg}} \left(x_{ij}\right) \le y_j, \forall j$$

$$\left.\begin{array}{l} \max_{\mathrm{i}} \{x_{ij}\} \le y_j \;\color{red}{\Rightarrow}\; \operatorname{avg}\left(x_{ij}\right) \le y_j \;\color{red}{\Rightarrow}\; P_{rel1} \subseteq P_{rel2} \\ \\ \operatorname{avg}\left(x_{ij}\right) \le y_j \quad \text{does not imply} \quad \max_{\mathrm{i}} \{x_{ij}\} \le y_j \;\Rightarrow\; P_{rel1} \ne P_{rel2} \end{array}\right\} \Rightarrow P_{rel1} \subset P_{rel2}$$

$\blacksquare$

- **Ideal formulations**

What is the best possible formulation?

MIP

$$\min \; c^T x$$
$$\text{s.t.} \quad Ax \le b$$
$$P_{ip} \longrightarrow \quad x_i \in Z, \;\; i \in \{1, \dots, n\}$$

Relaxation

$$\min \; c^T x$$
$$\text{s.t.} \quad Ax \le b \quad \longleftarrow \quad P_{rel}$$

Can you think of a relaxation whose optimal solution is the same as that of the MIP?

For such a relaxation we would have: $p^*_{rel} = p^*$

Does such a relaxation exist?

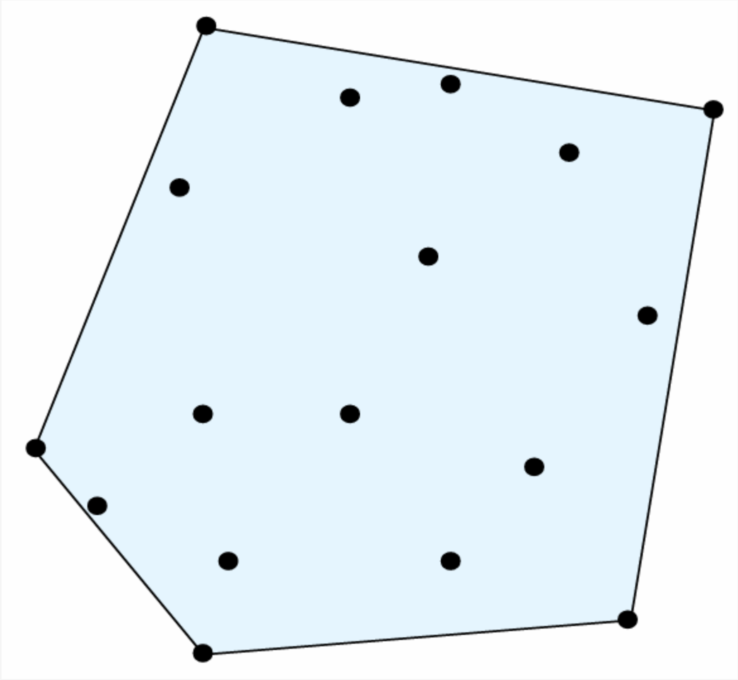
Convex hull

Refresher…

Given a set of points $C = \{x_1, x_2, \dots, x_n\}$ with $x_i \in R^n$.

The convex hull is the set of all possible convex combinations of the points in C

$$conv\ C = \left\{ \sum \alpha_i x_i \mid \mathbf{1}^T \alpha = 1, \alpha_i \geq 0, \forall i \right\}$$



What is the convex hull of an Integer Optimization problem?

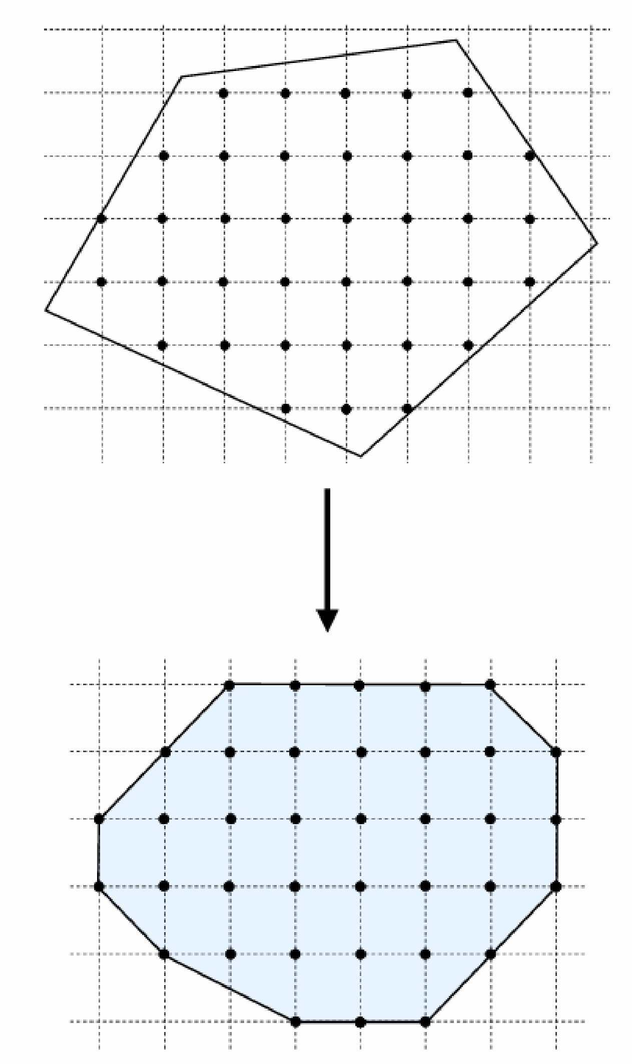
Convex hull of integer optimization

MIP

$$\min \; c^T x$$

$$\text{s.t.} \quad Ax \le b$$

$$P_{ip} \longrightarrow \quad x_i \in Z, \; i \in I \subseteq \{1, \dots, n\}$$



The convex hull has integer feasible extreme points

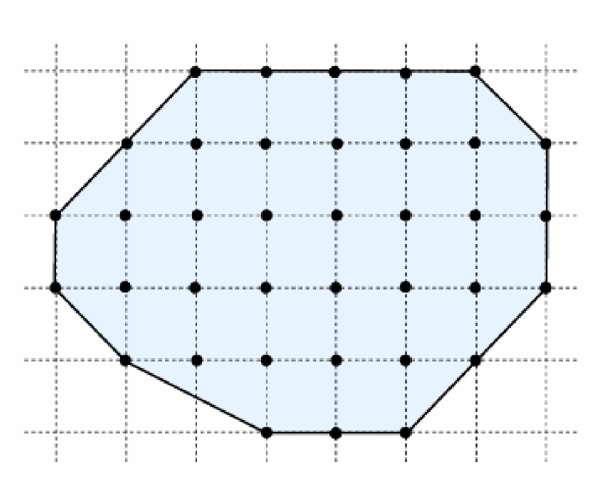
$$conv\, P = conv\{x \mid Ax \le b, \quad x_i \in Z, \quad i \in I\}$$

Ideal formulations

A formulation is called **ideal formulation** if solving its relaxation gives an integer feasible solution

$$\begin{aligned} \min\ & c^T x \\ \text{s.t.}\ & \boxed{\begin{aligned} & Ax \leq b \\ & x_i \in Z,\ i \in I \subseteq \{1, \dots, n\} \end{aligned}} \end{aligned} \longrightarrow \begin{aligned} \min\ & c^T x \\ \text{s.t.}\ & Ax \leq b \end{aligned} \longrightarrow \begin{matrix} \textit{Integer feasible} \\ x^* \end{matrix}$$

This happens when $conv\ P = \{Ax \leq b\}$



It is very hard to construct ideal formulations!

Facility location problem

Formulation 1:

$$\min \sum_{j=1}^{n} c_j y_j + \sum_{i=1}^{m} \sum_{j=1}^{n} d_{ij} x_{ij}$$

$$\text{s.t. } \sum_{j=1}^{n} x_{ij} = 1, \quad i = 1, \dots, m$$

$$x_{ij} \le y_j, \quad i = 1, \dots, m,\ j = 1, \dots, n$$

$$x_{ij}, y_j \in \{0,1\}, \ \forall i, j$$

Formulation 2 (fewer constraints):

$$\min \sum_{j=1}^{n} c_j y_j + \sum_{i=1}^{m} \sum_{j=1}^{n} d_{ij} x_{ij}$$

$$\text{s.t. } \sum_{j=1}^{n} x_{ij} = 1, \quad i = 1, \dots, m$$

$$\sum_{i=1}^{m} x_{ij} \le m y_j, \quad j = 1, \dots, n$$

$$x_{ij}, y_j \in \{0,1\}, \ \forall i, j$$

Relaxation rankings

$$conv\ P \subseteq P_{rel1} \subseteq P_{rel2}$$

Judging formulations

Size of feasible region

Goal: $conv\ P \approx \{Ax \leq b\}$

Objective function value

Goal: $p^*_{rel} \approx p^*_{ip}$

Problem size

Goal: keep moderate LP relaxation size

Problem formulations

$$\min\ c^T x$$

$$\text{s.t.}\quad Ax \leq b$$

$$x_i \in Z,\ \ i \in \{1, \dots, n\}$$

Remark: unfortunately, better formulations tend to have more constraints and/or variables

Minimum flow network flow

$$\begin{aligned} \min\ & c^T x \\ \text{s.t.}\ & Ax \le b \\ & 0 \le x \le u \end{aligned}$$

$\longrightarrow$

Integrality theorem: if A is totally unimodular (e.g., a graph arc-node incidence matrix), and b, u are integral vectors, then x^* is integral solution

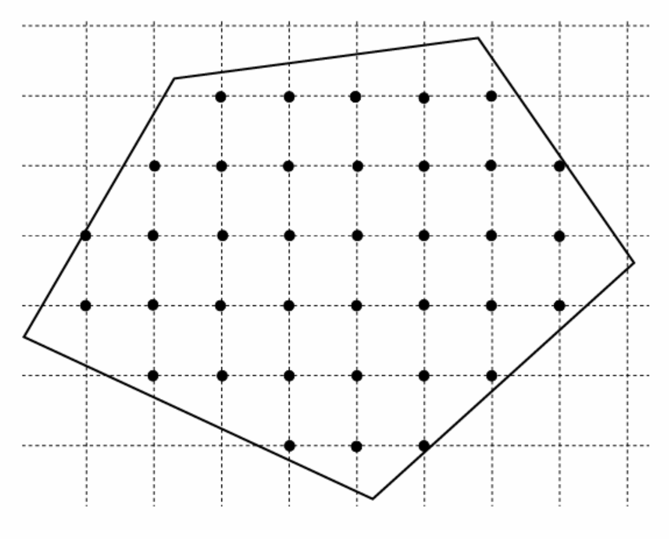
The formulation is ideal

$$\begin{aligned} \min\ & c^T x \\ \text{s.t.}\ & Ax \le b \\ & 0 \le x \le u \\ & \boldsymbol{x} \in \boldsymbol{Z}^{\boldsymbol{n}} \end{aligned}$$

Very easy special case!

How do we solve integer optimization problems

$$\min \; c^T x$$
$$\text{s.t.} \quad Ax \le b$$
$$x_i \in Z^n, \quad i \in I$$



Main idea: Refine the feasible set until the relaxation gives integer feasible solutions.

More details in the next lecture.

References

- D. Bertsimas & J. Tsitsiklis:
  - Ch. 10: integer programming formulations
- R. Vanderbei:
  - Ch. 23: integer programming

Next lecture

- Integer optimization solution algorithms